

RE05

Infrastructure of Rajasthan: Energy, Transport, Communication

Rajasthan Economy runs at 11-15 questions a paper and infrastructure supplies 2-4 of them. Two blocks are near-certain: renewable energy (asked every year since 2016) and thermal-station-to-district matching (2016, 2018, 2023). Scheme-to-year matching for UDAY, Saubhagya, PM-KUSUM and PM Surya Ghar is the third standing favourite.

This chapter is where Rajasthan's geography turns into marks. A desert state with the country's best sunshine has become India's number one solar state, and RPSC has noticed. Solar and renewable energy has appeared in every single paper since 2016. Around it sit the old reliables - which thermal station is in which district, who generates and who transmits, and where the Delhi-Mumbai Expressway cuts across the state.

Infrastructure of Rajasthan

Power institutions

- RSEB 1957 - unbundled 2000
- RVUNL = generation
- RVPNL = transmission
- 3 discoms: Jaipur, Ajmer, Jodhpur VVNL
- RERC (1999 Act) = tariff regulator
- RRECL = renewable nodal agency

Thermal & lignite

- Suratgarh (Sriganganagar) - largest, 2,820 MW
- Kota (Kota) - first coal station
- Chhabra (Baran)
- Kalisindh (Jhalawar)
- Giral & Jalipa-Kapurdi (Barmer)
- Barsingsar (Bikaner)

Nuclear, gas, hydel

- Rawatbhata (Chittorgarh) - only nuclear, PHWR
- Ramgarh gas (Jaisalmer)
- Dholpur combined cycle (Dholpur)
- Rana Pratap Sagar 172 MW
- Jawahar Sagar 99 MW, Mahi
- Bajaj Sagar 140 MW
- Chambal projects shared 50:50 with MP

Renewables - the exam core

- 1st in solar, 2nd in total RE (behind Gujarat)
- Bhadla Solar Park, Phalodi - India's largest
- Solar Energy Policy 2019 - 30 GW
- Green Hydrogen Policy 2023 - 2,000 KTPA
- Integrated Clean Energy Policy 2024 - 125 GW
- PM-KUSUM, PM Surya Ghar, Kisan Mitra Urja

Reform schemes

- UDAY 2015 - Rajasthan third to sign
- RDSS 2021 - smart prepaid meters
- DDUGJY 2015 - feeder separation
- Saubhagya 2017 - household connections

Transport

- Delhi-Mumbai Expressway NE-4 - 373 km in Rajasthan
- Amritsar-Jamnagar corridor
- PMGSY 2000; Mukhyamantri Gram Sadak Yojana
- RSRTC 1964, RSRDC 1979
- North Western Railway, HQ Jaipur, 2002
- Jaipur Metro 2015; Jaipur only international airport

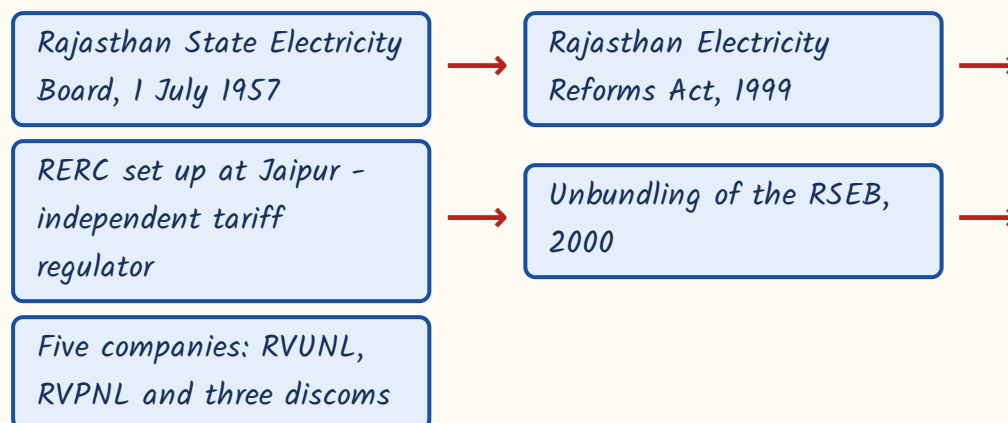
Digital & water

- BharatNet - fibre to gram panchayats
- RajSWAN + State Data Centre
- Jan Soochna Portal 2019 - India's first
- Jal Jeevan Mission - SS lpcd, 50:50 for Rajasthan
- RUIDP - ADB assisted urban infrastructure

The power sector's family tree - who generates, who carries, who bills

Before 2000 one body did everything. The Rajasthan State Electricity Board, constituted on 1 July 1957, generated power, carried it and sold it. That single-board model collapsed under losses everywhere in India. Rajasthan legislated its way out with the Rajasthan Electricity Reforms Act, 1999, set up an independent regulator, and then unbundled the Board in 2000 into five successor companies. Learn the split by function - RPSC asks it as a matching question.

How the RSEB became five companies



Who does what in Rajasthan's power sector

Body	Function	Note
RVUNL	Generation	Rajasthan Rajya Vidyut Utpadan Nigam
RVPNL	Intra-state transmission	Rajasthan Rajya Vidyut Prasaran Nigam
Jaipur / Ajmer / Jodhpur VVNL	Distribution	Exactly three discoms, all incorporated 2000
RERC, Jaipur	Tariff determination, licensing	Set up under the 1999 state Act

Body	Function	Note
RRECL, Jaipur	Renewable energy promotion	Nodal agency; successor to REDA

- Rajasthan's total installed generation capacity was 54.77 GW on 31 October 2025 (Rajasthan Energy Department).
- It was 22.02 GW in March 2020 - the state has more than doubled capacity in five years.
- Almost the whole of that addition is solar and wind, not coal.
- RERC continues to function under the central Electricity Act, 2003, but it was born under the state Act of 1999.
- RRECL is the successor to the Rajasthan Energy Development Agency (REDA).

TRAP

Two favourite traps here. First - there is no Kota discom. Rajasthan has exactly three: Jaipur, Ajmer and Jodhpur Vidyut Vitran Nigam. Second - RERC was created by the Rajasthan Electricity Reforms Act 1999, NOT by the Electricity Act 2003. The option saying 2003 is put there deliberately.

Thermal power stations - the district-matching question RPSC

keeps asking

This is pure memory work, and it pays. RPSC asked station-to-district matching in 2016, 2018 and 2023. Fix five pairs and you have the question. Suratgarh is the biggest and carries a nickname; Kota is the oldest. Do not mix those two up.

Coal-based thermal stations

Station	District	Capacity / feature
Suratgarh Super Thermal	Sriganganagar	2,820 MW; two 660 MW supercritical units; largest in the state
Kota Super Thermal	Kota (on the Chambal)	1,240 MW; Rajasthan's FIRST coal-based station; repeated national productivity awards
Chhabra Thermal	Baran	1,000 MW subcritical + 1,320 MW supercritical; coal, not gas
Kalisindh Thermal	Jhalawar	2 x 600 MW supercritical

- Suratgarh is called the 'Thermal Power House of Rajasthan' - a straight one-liner asked in 2016.
- Kota Super Thermal is Rajasthan's first coal-based station and stands on the bank of the Chambal.
- Chhabra and Kalisindh are the two supercritical projects of the Hadoti belt - Baran and Jhalawar respectively.
- Suratgarh sits in the Indira Gandhi canal command in the far north-west; Kota, Chhabra and Kalisindh are all in the south-east.

ASKED IN RAS

Asked as a match-the-following in RAS Pre 2018 and again in 2023 - Suratgarh to Sriganganagar, Chhabra to Baran, Kalisindh to Jhalawar, Giral to Barmer. Four pairs, one question, done.

TRAP

Do not confuse the two superlatives. Suratgarh = LARGEST thermal station. Kota = FIRST coal-based station. And Chhabra (Baran) is coal-based - the paper offers 'natural gas' against it to see if you are guessing.

Lignite, gas, hydel and the one nuclear station

Rajasthan burns its own lignite in the west, its own gas in Jaisalmer and Dholpur, and takes hydel power from the Chambal and Mahi. There is exactly one nuclear station. Every one of these is a district-matching fact, so learn the place along with the plant.

- Lignite stations: Giral (Barmer, 250 MW), Barsingsar (Bikaner, run by NLC India), and the Bhadresh station in Barmer fed by the Jalipa-Kapurdi lignite fields.
- Gas-based: Ramgarh (Jaisalmer) and the Dholpur Combined Cycle plant (Dholpur, 330 MW).
- Jalipa and Kapurdi are lignite MINES, not power stations - the station they feed is in Barmer district.
- Barmer therefore carries two lignite stations, Giral and Bhadresh.

Hydel projects to districts

Project	District	Capacity
Rana Pratap Sagar	Chittorgarh	172 MW
Jawahar Sagar	Kota	99 MW
Mahi Bajaj Sagar	Banswara	140 MW

Rawatbhata Atomic Power Station is Rajasthan's only nuclear station. It stands in Chittorgarh district, on the Chambal near the Rana Pratap Sagar dam. Unit-1 was built with Canadian collaboration and used pressurised heavy water reactor technology - the PHWR line India went on to indigenise. Remember also that the Chambal valley projects are shared 50:50 in cost and power between Rajasthan and Madhya Pradesh.

YAAD RAKHO

Trick for the hydel trio: Rana Pratap belongs to Mewar, so Chittorgarh.

Jawahar Sagar is downstream at Kota. Mahi is the Vagad river, so Banswara.

Three rulers, three regions, three districts.

Renewables - RPSC has asked this in every paper since 2016

This is the single highest-yield section in the chapter. Rajasthan has the country's best solar radiation and the country's largest stock of cheap barren land, and it has converted both into capacity. State the position precisely, because the paper tests precision: Rajasthan is **FIRST** among states in installed solar capacity, but **SECOND** in total renewable capacity, behind Gujarat, as of March 2026. Both halves of that sentence get asked.

- Rajasthan ranks first in installed solar capacity - about 36 GW, roughly 27 per cent of India's cumulative solar (MNRE, as on 30 November 2025).
- Gujarat was second in solar with 24.8 GW and Maharashtra third with 17.2 GW on the same date.
- In **TOTAL** renewable capacity, Gujarat leads with 47,178 MW against Rajasthan's 47,021 MW (as on 31 March 2026) - a gap of under 200 MW.
- Rajasthan and Gujarat together carry over one-third of India's renewable capacity.
- Rajasthan's wind capacity is concentrated in the western desert districts - Jaisalmer, Jodhpur and Barmer - and in Deogarh; not in Hadoti.
- RRECL, Jaipur is the state **NODAL AGENCY** for solar, wind, biomass and green hydrogen.

Rajasthan vs Gujarat - learn which rank is which

Rajasthan	Gujarat
<i>FIRST in installed solar capacity</i>	<i>Second in solar</i>
<i>About 36 GW solar (30 Nov 2025)</i>	<i>About 24.8 GW solar (30 Nov 2025)</i>
<i>SECOND in total renewable capacity</i>	<i>First in total renewable capacity</i>
<i>47,021 MW total RE (31 Mar 2026)</i>	<i>47,178 MW total RE (31 Mar 2026)</i>
<i>Strength: solar, on desert wasteland</i>	<i>Strength: wind plus solar, long coastline</i>

ASKED IN RAS

Asked in RAS Pre 2018 - which is the nodal agency for renewable energy in Rajasthan? Answer: Rajasthan Renewable Energy Corporation Limited. Asked again in 2023 in the form 'which state ranks first in installed solar capacity' - Rajasthan.

TRAP

Bhadla Solar Park is INDIA's largest solar park, not the world's. Older coaching notes still say 'world's largest' - do not carry that into the exam hall. Also note the district: Bhadla village is now in Phalodi district, carved out of the old Jodhpur district, so both 'Jodhpur' and 'Phalodi' appear in older and newer sources.

CURRENT LINK

Current link, as of September 2026: Rajasthan's installed capacity has crossed 54.77 GW (31 October 2025) from 22.02 GW in March 2020, and the state has moved to within 200 MW of Gujarat in total renewable capacity (March 2026 data). Any question on 'first in solar' or 'second in renewables' will be built on these numbers - check the MNRE monthly state-wise sheet once more in November 2026.

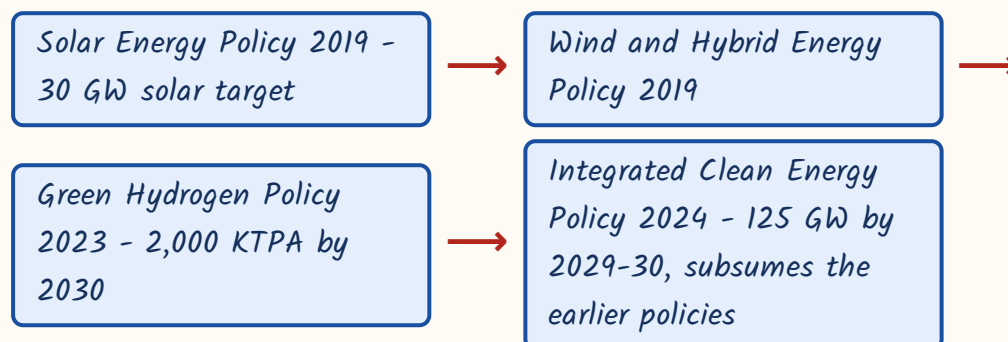
Bhadla, the clean energy policies and the three schemes to

memorise

Bhadla Solar Park is the flagship. It lies at Bhadla village in Phalodi district, on land so hot and barren that nothing else could be done with it. Capacity is about 2,245 MW spread over roughly 5,700 hectares, developed in four phases and completed by March 2020. Learn the capacity figure - it was asked directly in 2021.

- Bhadla Solar Park: about 2,245 MW, roughly 5,700 hectares, four phases, completed March 2020.
- Location: Bhadla village, Phalodi district (carved out of the erstwhile Jodhpur district).
- It is India's largest solar park by installed capacity.
- Bhadla was NOT commissioned before 2010 - construction ran from 2015 to 2020. That false statement is a standard distractor.

How Rajasthan's clean energy policy evolved



Policy to year to target

Policy	Year	Target / feature
Rajasthan Solar Energy Policy	2019	30 GW solar

Policy	Year	Target / feature
Rajasthan Wind and Hybrid Energy Policy	2019	Wind and wind-solar hybrid projects
Rajasthan Green Hydrogen Policy	2023 (approved September)	2,000 KTPA by 2030; exemption from wheeling and transmission charges; thrust sector under RIPS
Rajasthan Integrated Clean Energy Policy	2024	125 GW by 2029-30 = 90 GW solar + 25 GW wind and hybrid + 10 GW hydro, pumped storage and battery storage

- PM-KUSUM (2019) is run by the Ministry of New and Renewable Energy, not by an agriculture ministry - a favourite catch.
- PM-KUSUM Component-A: decentralised ground-mounted renewable plants up to 2 MW on barren land. Rajasthan leads the country in Component-A.
- Component-B: standalone solar agricultural pumps. Component-C: solarisation of grid-connected pumps.
- PM Surya Ghar: Muft Bijli Yojana - launched 13 February 2024, outlay about Rs 75,021 crore, rooftop solar for one crore households, up to 300 free units a month.
- Mukhyamantri Kisan Mitra Urja Yojana (2021, Rajasthan Energy Department) - grant of up to Rs 1,000 a month, maximum Rs 12,000 a year, credited against the bills of METERED agricultural consumers.

ASKED IN RAS

Three separate asks from this block. RAS Pre 2021 - Bhadla's capacity (2,245 MW) and the Solar Energy Policy 2019 target (30 GW). RAS Pre 2024 - free units under PM Surya Ghar (300 a month). RAS Pre 2023 - the Kisan Mitra Urja ceiling (Rs 1,000 a month, Rs 12,000 a year). A 2021 statement question also tested all three PM-KUSUM components together.

Discom finances and getting power to the last house

Generation was never Rajasthan's problem. Distribution was. The three discoms ran up the largest debt in the country, which is why UDAY matters here more than in most states. Understand the logic in one line: the state takes over most of the discom debt, and in return the discoms must cut their losses to an agreed level.

- UDAY - Ujwal DISCOM Assurance Yojana, launched 20 November 2015 by the Ministry of Power.
- States were to take over 75 per cent of discom debt - 50 per cent in 2015-16 and 25 per cent in 2016-17.
- Target for aggregate technical and commercial (AT&C) losses under UDAY: about 15 per cent.
- Rajasthan was the THIRD state to sign a UDAY memorandum of understanding, in December 2015, and carried the largest discom debt in the country.
- Revamped Distribution Sector Scheme (RDSS), 2021 - outlay about Rs 3.03 lakh crore for prepaid smart metering and loss-reduction works, targeting AT&C losses of 12-15 per cent.
- RDSS is a distribution scheme. It has nothing to do with building hydro projects - that pairing appears as a wrong option.
- Deen Dayal Upadhyaya Gram Jyoti Yojana (2015) - rural feeder separation and rural distribution strengthening.
- Saubhagya, officially the Pradhan Mantri Sahaj Bijli Har Ghar Yojana (September 2017) - free electricity connections to willing un-electrified households.

- DDUGJY works on the WIRES; Saubhagya works on the CONNECTION inside the house. That is the distinction the paper tests.

YAAD RAKHO

Yaad rakho the energy scheme ladder in order: UDAY 2015, Saubhagya 2017, PM-KUSUM 2019, RDSS 2021, PM Surya Ghar 2024. RAS asked exactly this as a chronological ordering question. Odd years all the way to 2021, then 2024.

Roads – length, agencies and the expressway that redrew the map

Rajasthan is India's largest state by area, so road length is large but road density is only moderate. Learn the agencies by their sphere of work, because RPSC matches organisation to function rather than asking bare kilometres.

- Total road length in Rajasthan was about 2,64,244 km with a road density of 77.21 km per 100 sq km (as on March 2019) – quote the date, since these figures are revised.
- The Public Works Department maintains the bulk of it – over 2.01 lakh km.
- National highway length in the state was about 7,906 km (as on 30 June 2017) and has grown since; use the figure with its date.
- RIDCOR runs the Mega Highways Project, begun in 2005, on a public-private partnership model.

Transport and infrastructure agencies

Agency	Established	Sphere of work
RSRTC	1 October 1964, HQ Jaipur	Passenger road transport; set up under the Road Transport Corporations Act, 1950
RSRDC	1979	State road and bridge construction on a corporate model

Agency	Established	Sphere of work
RUIDP	2000	ADB-assisted urban water supply, sewerage, drainage and roads
RRECL	-	Renewable energy promotion (nodal agency)

The Delhi-Mumbai Expressway, designated NE-4, is the biggest single change to Rajasthan's transport map in decades. It is about 1,350 km long, eight lanes wide and expandable to twelve. Rajasthan carries 373 km of it - the **SECOND** longest state share, after Gujarat's 426 km. Alongside it, the Amritsar-Jamnagar Economic Corridor gives western Rajasthan a freight road to the Kandla and Jamnagar ports.

- Delhi-Mumbai Expressway in Rajasthan passes through Khairthal-Tijara/Alwar, Dausa, Sawai Madhopur, Bundi and Kota.
- The first stretch opened was the 246 km Sohna-Dausa-Lalsot section, on 12 February 2023.
- State shares: Gujarat 426 km (longest), Rajasthan 373 km, Madhya Pradesh 244 km, Haryana 129 km.
- The Amritsar-Jamnagar corridor enters Rajasthan at Hanumangarh and runs through Bikaner, Jodhpur, Barmer and Jalore.
- PMGSY (25 December 2000, Ministry of Rural Development) connects habitations of 500 and above - relaxed to 250 and above in desert, Schedule V tribal and hill areas.
- Rajasthan runs its own Mukhyamantri Gram Sadak Yojana alongside PMGSY.

Railways, the metro and the airports

One railway zone is headquartered in Rajasthan and one is not - that distinction is the whole question. Learn the four divisions of the North Western

Railway and remember that Kota is not one of them.

- North Western Railway zone, headquarters Jaipur, created on 1 October 2002.
- Its four divisions: Jaipur, Ajmer, Bikaner and Jodhpur.
- Kota division belongs to the West Central Railway zone, not the NWR - a standard 'which is NOT' option.
- The Western Dedicated Freight Corridor runs from Jawaharlal Nehru Port Trust to Dadri; the Rewari-Madar (Ajmer) section through Rajasthan was among the first commissioned.
- Jaipur Metro is Rajasthan's first metro rail. Phase-1A, Mansarovar to Chandpole, opened on 3 June 2015.
- Jaipur (Sanganer) is Rajasthan's ONLY international airport.
- Kishangarh airport, Ajmer district, was commissioned in 2017 under the UDAN regional connectivity scheme - it is a domestic airport.
- Other operating airports: Jodhpur, Udaipur (Maharana Pratap, Dabok), Jaisalmer and Bikaner (Nal).
- Rajasthan has no coastline, so dry ports and inland container depots do the work of ports - the main ICDs serve Jaipur, Jodhpur, Bhilwara and Bhiwadi.
- The state's freight logic is simple: DFC and the Amritsar-Jamnagar corridor carry goods west to Gujarat's ports; NE-4 carries them south-west to Mumbai.

Digital infrastructure, and the water and sanitation backbone

Rajasthan's e-governance reputation rests on two things - the physical network that carries the data, and one portal that made the state first in the country. Both are asked.

- BharatNet, originally the National Optical Fibre Network, takes optical fibre up to the GRAM PANCHAYAT level; later phases extend it to villages.
- RajSWAN, the Rajasthan State Wide Area Network, plus the Rajasthan State Data Centre, provide connectivity and hosting for state e-governance.
- They carry the platforms you must name: e-Mitra, Jan Aadhaar, Raj Single Sign-On and Rajasthan Sampark.
- Jan Soochna Portal - launched 13 September 2019, India's FIRST portal for proactive disclosure of government information under Section 4(2) of the RTI Act. Asked in RAS Pre 2021.

On the water side, two names carry the chapter. The Jal Jeevan Mission (15 August 2019) promises a functional household tap connection giving 55 litres per capita per day of water meeting BIS 10500 quality. For a general-category state like Rajasthan the Centre-State sharing is 50:50 - not 90:10. RUIDP is the Asian Development Bank-assisted urban infrastructure programme running since 2000 for town water supply, sewerage, drainage and roads.

- JJM service level: 55 lpcd, BIS 10500 quality, through a functional household tap connection.
- Village Water and Sanitation Committees and gram panchayats operate and maintain the schemes.
- JJM 2.0 was approved in March 2026 with an outlay of about Rs 8.69 lakh crore and a target date of December 2028.
- Nationally about 15.72 crore rural households had tap connections by March 2026 - check the Rajasthan state figure on the JJM dashboard closer to the exam, as it moves every month.
- Rural drinking water in Rajasthan is executed by the Public Health Engineering Department (PHED).

REVISION RECAP

1. RSEB constituted 1 July 1957; unbundled in 2000 into five companies - RVUNL, RVPNL and three discoms.
2. Discoms are exactly three: Jaipur, Ajmer, Jodhpur VVNL. No Kota discom.
3. RERC, Jaipur - created by the Rajasthan Electricity Reforms Act 1999, not the Electricity Act 2003.
4. Suratgarh (Sriganganagar), 2,820 MW - largest thermal, 'Thermal Power House of Rajasthan'.
5. Kota Super Thermal (1,240 MW) - Rajasthan's first coal-based station, on the Chambal.
6. Chhabra = Baran (coal). Kalisindh = Jhalawar. Giral = Barmer (lignite). Barsingsar = Bikaner (lignite).
7. Jalipa-Kapurdi lignite fields feed the Bhadresh station in Barmer district.
8. Ramgarh gas = Jaisalmer; Dholpur combined cycle = Dholpur (330 MW).
9. Rawatbhata, Chittorgarh - only nuclear station; Unit-I Canadian collaboration, PHWR.
10. Rana Pratap Sagar 172 MW (Chittorgarh), Jawahar Sagar 99 MW (Kota), Mahi Bajaj Sagar 140 MW (Banswara); Chambal shared 50:50 with MP.
11. Rajasthan: FIRST in solar (about 36 GW, 27 per cent of India, 30 Nov 2025), SECOND in total renewables behind Gujarat (March 2026).
12. RRECL is the renewable nodal agency; total installed capacity 54.77 GW on 31 October 2025.
13. Bhadla Solar Park - Phalodi district, about 2,245 MW, 5,700 hectares, four phases to March 2020, India's largest.
14. Policies: Solar 2019 (30 GW), Wind & Hybrid 2019, Green Hydrogen 2023 (2,000 KTPA), Integrated Clean Energy 2024 (125 GW by 2029-30).
15. PM-KUSUM is with MNRE: A = plants up to 2 MW, B = standalone pumps, C = grid-connected pump solarisation; Rajasthan leads Component-A.
16. PM Surya Ghar (13 Feb 2024) - 300 free units a month, one crore households, about Rs 75,021 crore. Kisan Mitra Urja - Rs 1,000 a month, Rs 12,000 a year, metered farm consumers.
17. UDAY (20 Nov 2015) - 75 per cent debt takeover, 15 per cent AT&C target; Rajasthan the third signatory. RDSS 2021 - smart prepaid metering, 12-15 per cent target.
18. Delhi-Mumbai Expressway = NE-4, about 1,350 km; Rajasthan 373 km (second after Gujarat's 426 km); first stretch Sohna-Dausa-Lalsot, 12 February 2023.
19. RSRTC 1964 (buses), RSRDC 1979 (roads and bridges), RUIDP 2000 (ADB urban). PMGSY 2000 - 500+, relaxed to 250+ in desert, tribal and hill areas.
20. North Western Railway, HQ Jaipur, 2002, four divisions (Jaipur, Ajmer, Bikaner, Jodhpur - not Kota); Jaipur Metro 3 June 2015; Jaipur the only international airport; Jan Sookhna Portal 2019, India's first.